Quasi Closed Phase Glottal Inverse Filtering Analysis With Weighted Linear Prediction

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***Abstract*—This study presents a new glottal inverse filtering (GIF) technique based on closed phase analysis over multiple fundamental periods. The proposed quasi closed phase (QCP) analysis method utilizes weighted linear prediction (WLP) with a specific attenuated main excitation (AME) weight function that attenuates the contribution of the glottal source in the linear prediction model optimization. This enables the use of the autocorrelation criterion in linear prediction in contrast to the covariance criterion used in conventional closed phase analysis. The QCP method was compared to previously developed methods by using synthetic vowels produced with the conventional source-filter model as well as with a physical modeling approach. The obtained objective measures show that the QCP method improves the GIF performance in terms of errors in typical glottal source parametrizations for both low- and high-pitched vowels. Additionally, QCP was tested in a physiologically oriented vocoder, where the analysis/synthesis quality was evaluated with a subjective listening test indicating improved perceived quality for normal speaking style.**

***Index Terms*—Closed phase analysis, GIF, glottal inverse filtering, speech analysis, weighted linear prediction.**

I. Introduction

A WIDELY used way to categorize different speech sounds is to divide them into three groups (*voiced*, *unvoiced*, and *plosives*) according to the functioning of the human voice production mechanism [1]. In each group, a speech signal involves an *excitation* and a vocal tract *filter*. Voiced sounds are excited by the periodic flow of air—the *glottal volume velocity waveform*—generated by the vibration of the vocal folds. Unvoiced sounds are excited in the absence of vocal fold vibration by turbulent noise. Plosives, which can be either voiced or unvoiced, are excited by a transient-type abrupt release of air flow. The acoustic speech pressure signal is formed by filtering the excitation through the vocal tract and by taking into account the *lip radiation* (i.e., conversion of the air flow at the lips into a pressure signal outside the mouth).

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The glottal volume velocity waveform, or in shorter terms the glottal flow, can be estimated with glottal inverse filtering (GIF), a technique that is based on the use of properly adjusted antiresonances that cancel the effects of the vocal tract and lip radiation from a recorded speech signal. GIF benefits from the fact that the analysis is non-intrusive, it can be performed solely based on the speech sound pressure signal, and it can be implemented to perform in a completely automatic manner.

Various GIF methods have been proposed, starting from the 1950s (see [2] for a historical review of GIF). The earliest methods utilized manually-tuned analog antiresonance circuits, but starting from the 1970s the use of digital processing and filtering has been the basis of GIF techniques.

Closed phase (CP) covariance analysis [3] is one of the most well known digital automatic GIF methods. In the CP method, the vocal tract filter is estimated by computing the linear prediction (LP) analysis with the covariance criterion over speech samples that are located in the closed phase of the glottal cycle. Another prevalent approach for GIF is to estimate the contribution of the glottal flow with a low-order LP analysis. One of the most common methods of this type is Iterative Adaptive Inverse Filtering (IAIF) [4] that utilizes an iterative scheme of estimating in turn the glottal flow and the vocal tract spectral envelopes. Homomorphic analysis [5], [6] has also been proposed and successfully applied in GIF [7]–[10]. The original Zeros of the Z-Transform (ZZT) method, and the more recent, computationally more effective Complex Cepstrum Decomposition (CCD) method separate the contributions of the glottal source and the vocal tract based on decomposing speech into minimum-phase and maximum-phase components.

Several previous studies have indicated that the accuracy of GIF methods is, in general, sufficient in estimating the glottal flow from low-pitched male voices [3], [11]. However, when the fundamental frequency (F_0) of speech increases the performance of GIF methods generally deteriorates [2]. In addition to F_0, the type of phonation (i.e., modal, breathy, pressed) affects the accuracy of GIF (depending on the method). It is also worth emphasizing that the evaluation of the GIF accuracy is problematic from the following principal point of view: since the true glottal flow signal cannot be measured directly from the human larynx, there is no ground truth available in GIF evaluations. Most studies have therefore used synthetic speech waveforms to obtain objectively verifiable measures to evaluate GIF methods. Synthetic test utterances are usually generated by using a parametric glottal flow pulseform, such as the Liljencrants-Fant (LF) model [12] as an excitation to an all-pole filter that is adjusted to model the formant structure of a given vowel.

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A new GIF algorithm is presented in this study. The method, quasi closed phase (QCP) analysis, is based on the principles of the CP analysis, that is, the estimation of the vocal tract during the closed phase of glottal flow. Differently from the CP method, the proposed technique does not compute the vocal tract filter with the covariance method from few samples located in the closed phase. Instead, QCP analysis takes advantage of *weighted linear prediction* (WLP) in order to utilize all the data samples of the analysis frame of multiple pitch periods by putting more emphasis on the samples which are located in the closed phase. It is hypothesized that this makes the analysis more robust with respect to the detection of the closed phase.

The QCP method was introduced in a conference paper in [13]. The current study is a sequel to [13] providing extended evaluations and analyses of the proposed method. Evaluation is conducted by utilizing test vowels synthesized both by the conventional source-filter model as well as by a physical modeling approach. The evaluation involves comparing the proposed method with three known techniques: CP, IAIF, and CCD. The properties of the proposed method are analyzed by examining its robustness to errors in glottal closure instant detection. The study is partly motivated by the increasing interest of the speech synthesis community in vocoding techniques utilizing GIF [14], [15]. In order to take this potential application area into account, the investigation also involves tests on the perceived analysis/synthesis quality of a vocoder that uses GIF as a basis for speech parametrization.

This article is arranged as follows: In Section II, the fundamental theory behind speech production and modeling relevant to the proposed method is briefly reviewed. In Section III, the WLP all-pole modeling technique is described. In Section IV, the QCP analysis method is described, and the details of its parameter selections are presented in Section IV-A. In Section V, four reference methods that are compared to QCP analysis are described in detail. In Section VI, the conducted tests with the proposed method are introduced, and their results are presented in Section VII. In Section VIII, the conclusions of the study are presented.

II. The Source-Filter Model

The glottal flow waveform is produced by the air flow streaming from the lungs generating oscillations at the vocal folds. Within each cycle, the vocal folds vibrate from an open position into a closed position, and these phases constitute the two main parts of the glottal flow pulse: the open phase and the closed phase (Fig. 1). During the open phase, the air flow from the lungs first gradually increases, until it comes to an abrupt stop at the glottal closure instant (GCI). GCIs can be observed particularly well in the glottal flow derivative waveform illustrated in Fig. 1(b), where the negative peak corresponds to the largest deceleration in glottal volume velocity. During the closed phase, there is no air flow between the orifice of the vocal folds—the glottis—if the fluctuation mode is modal or pressed. In breathy phonation, however, there is a non-zero leakage component in the glottal flow during the entire duration of the glottal cycle.

By estimating the lip radiation effect as a first order differentiator, the glottal flow derivative serves as the effective driving

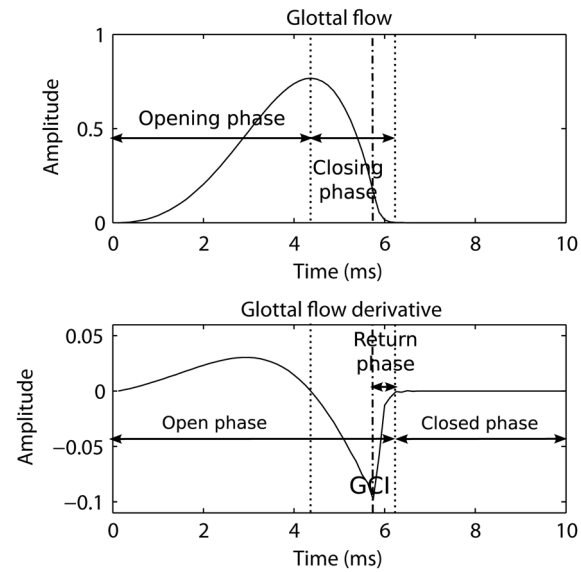

Fig. 1. The glottal flow waveform (upper graph), and the glottal flow derivative waveform (lower graph) as depicted by the LF model.

excitation $E(z)$ of the speech signal in the source-filter model of speech production [1]. In mathematical form this can be expressed in the z-domain as follows:

$$S(z) = G(z)V(z)L(z) = E(z)V(z), \tag{1}$$

where $S(z)$ is the speech signal, $G(z)$ is the glottal excitation, $V(z)$ is the vocal tract transfer function, and $L(z)$ is the transfer function of the lip radiation effect. As the transfer function of the lip radiation effect is usually assumed to be known and of the form

$$L(z) = 1 - \alpha z^{-1}, \tag{2}$$

where α is a constant within the range [0.96,1[, GIF methods are left with the task of accurately estimating the vocal tract transfer function $V(z)$ to obtain an estimate of $G(z)$.

III. Weighted Linear Prediction

WLP was first proposed by Ma *et al.* in [16]. It is one of several modified LP methods that have been proposed to provide all-pole models that are less affected by F_0 and its harmonics [17]–[19].

WLP uses a similar prediction model to conventional LP:

$$s_n = \sum_{i=1}^{p} s_{n-i} a_i + e_n, \tag{3}$$

where s_n denotes the n th sample of a speech wave, e_n is the n th sample of an excitation (residual) wave, a_i is the i th predictor coefficient, and p is the prediction order. WLP differs from conventional LP in the sense that it imposes *temporal weighting* on the square of the residual:

$$E = \sum_{n=n_1}^{n_2} \left(s_n - \sum_{i=1}^{p} s_{n-i} a_i \right)^2 W_n, \tag{4}$$

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where E is the total residual energy, $n_1 = 1$ and $n_2 = N + p$ for the autocorrelation case, N is the frame length (in samples), and W_n is a temporal weight function. It is important to note that W_n is *not* an ordinary time-domain windowing function (i.e., Hamming, Hann, etc.) that is widely used, for example, in the computation of autocorrelation terms in LP. Instead, W_n represents a temporal weight function with which the contribution of selected squared residual samples can be downgraded in the optimization of the filter coefficients. The parameters a_i are obtained by setting the partial derivatives of E with respect to a_i to zero, resulting in a set of p modified LP equations:

$$\sum_{k=1}^{p} a_k \sum_{n=n_1}^{n_2} W_n s_{n-k} s_{n-i} = \sum_{n=n_1}^{n_2} W_n s_n s_{n-i}, 1 \leq i \leq p, \tag{5}$$

which can be expressed in vector form as

$$\left(\sum_{n=n_1}^{n_2} W_n \cdot \mathbf{s}_n \mathbf{s}_n^T \right) \mathbf{a} = \sum_{n=n_1}^{n_2} W_n \cdot s_n \mathbf{s}_n, \tag{6}$$

where $\mathbf{a} = [a_1, a_2, \ldots, a_p]^{\mathrm{T}}$ and $\mathbf{s}_n = [s_{n-1}, s_{n-2}, \ldots, s_{n-p}]^{\mathrm{T}}$. Unlike in the traditional autocorrelation LP, the matrix $\mathbf{R} = \sum_{n=n_1}^{n_2} W_n \cdot \mathbf{s}_n \mathbf{s}_n^T$ is not a Toepliz matrix, which prohibits the use of the computationally efficient Durbin recursion [1] for the computation of the predictor coefficients $\mathbf{a}$. Instead, WLP computes a weighted autocorrelation matrix, and the resulting all-pole filter is not guaranteed to be stable [16]. In applications where stability of the all-pole filter needs to be guaranteed, it is possible to use Stabilized Weighted Linear Prediction (SWLP) [19]. SWLP modifies the weighted autocorrelation matrix in an iterative manner (for further details, see equations 8–11 in [19]), resulting in predictor polynomials which are always minimum phase.

A. The Weight Function

The selection of the weight function W_n for the WLP analysis is an essential part of the proposed method. In [16], a short-term energy (STE) function was proposed as the weight function for WLP analysis for the task of formant estimation:

$$W_n = \sum_{i=0}^{M-1} s_{n-1-i}^2, \tag{7}$$

where M denotes the length of the energy window. The advantage of the STE function is that it can be computed with ease straight from the input signal. An alternative weight function, attenuated main excitation (AME), was, however, recently proposed in [20]. The AME weight function, shown in Fig. 2, is based on forming a temporal waveform which downgrades the contribution of speech samples that are located in the vicinity of the main excitation of the vocal tract near the GCI. In [20], the AME function was used in WLP to estimate formants from high-pitched speech. It was shown that the AME weighting reduced the effect of the glottal source in the computation of WLP models. Consequently, more accurate formant estimates were obtained, especially for high-pitched speech.

The computation of the AME function is illustrated in Fig. 2 and described as follows: For each fundamental cycle, W_n is

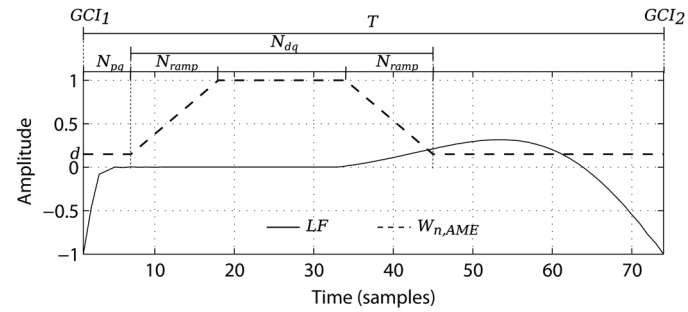

Fig. 2. A glottal flow derivative waveform (LF) and the AME weight function ($W_{n,\mathrm{AME}}$) between two consecutive GCIs. Glottal flow derivative was computed from the LF model. The AME function was computed with $PQ = 0.1$, $DQ = 0.5$, and $N_{ramp} = 10$.

equal to 1.0 in all time instants except in the vicinity of GCI where W_n becomes attenuated and equals a small positive constant, denoted by d ($d \approx 0, d > 0$). The value of d used in this article is 10^{-5}. The non-zero value for d is chosen to avoid concerns regarding the singularity of the weighted autocorrelation matrix. W_n changes smoothly between d and 1.0 by following a linear ramp whose length can be fixed to a constant of N_{ramp} samples or alternatively be set according to the pitch-proportional Ramp Quotient (RQ) which is defined as $RQ = N_{ramp}/T$. The position and length of the attenuated section are defined by two pitch proportional quantities, PQ (Position Quotient) and DQ (Duration Quotient). PQ, defined as $PQ = N_{pq}/T$, indicates the relative starting position of the non-attenuated section of W_n from the previous GCI within the glottal cycle. DQ, defined as $DQ = N_{dq}/T$, is a measure for the relative length of the non-attenuated section of W_n inside a glottal cycle.

IV. Quasi Closed Phase Analysis Method

The separation of the vocal tract transfer function from the effective driving excitation can be seen as a deconvolution problem where the effective driving excitation is strongest at short intervals in the vicinity of GCIs. During the closed phase, the coupling effect of the excitation to the speech signal is minimal [21]. It is assumed that the speech signal during these sections is mostly defined by the characteristic resonances of the vocal tract in free oscillating decay. Thus the spectral envelope analysis of the signal during the closed phase is expected to yield a vocal tract model that is relatively free from the contribution of the effective driving excitation. It is to be noted that this assumption is accurate only when the effective driving excitation is a Dirac delta pulse.

The justification above forms the basis for the CP covariance method, and it has been shown to yield good results when the estimation of the GCIs (and the closed phase) is accurate and the pitch of the signal is not too high [11]. Apart from the need for accurate GCI information, the problem with the CP analysis method is that the temporal duration of the closed phase decreases as the pitch of the voice increases. The limited number of samples on which the spectral envelope analysis is performed increases errors in vocal tract models especially when combined with the need for accurate GCI information. The use of the autocorrelation criterion in the LP analysis would result in stable

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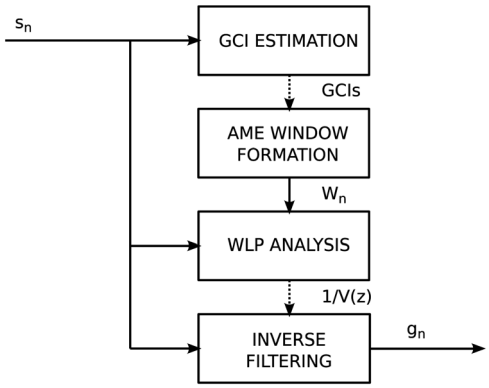

Fig. 3. Block diagram for the proposed QCP method.

and more consistent results, but the autocorrelation criterion requires significantly longer analysis frames than the covariance criterion, and as such the CP method is not applicable with the autocorrelation criterion.

In the proposed QCP method, the idea behind CP analysis—the estimation of the vocal tract during the closed phase—is applied with the autocorrelation criterion using a long analysis window. This is enabled by WLP in which the AME weight function (instead of the conventional STE weight function) is set to attenuate the effect of open phase samples in the LP optimization. Using this method, a common 20–50 ms speech frame can be inverse filtered simply by performing the appropriate WLP analysis and filtering the frame with the obtained FIR analysis filter, as depicted in the block diagram of Fig. 3. The algorithm consists of four main computational blocks. First, GCIs are detected from the speech signal to be later used in building the AME weight function. GCIs can be extracted, for example, by using electroglottography [22] or specific GCI detection algorithms such as DYPSA [23] or SEDREAMS [24], [25]. Second, the AME function is built as explained in Section III-A by using the estimated GCIs, from which the fundamental period T is also obtained. Third, the WLP based vocal tract model, denoted by $V(z)$ in Fig. 3, is estimated from the pre-emphasized speech signal according to Eq. (6) by using the AME function as W_n. Pre-emphasis ($\alpha = 0.99$) is used to approximately cancel out the lip radiation effect as discussed in Section II. On average, the AME function attenuates the effect of (quasi) open phase samples in the optimization of the vocal tract filter coefficients, thus performing a QCP analysis over multiple fundamental periods of the speech signal s_n in an analysis frame (similar idea was recently studied in [26] by using Stabilized Weighted Linear Prediction, SWLP [19]). Fourth, an estimate for the glottal flow derivative, denoted by g_n in Fig. 3, is obtained by inverse filtering the input speech signal with $V(z)$.

The QCP method is essentially the same as the proposed AME-WLP formant estimation method in [20] on a basic level. The difference between the two arise from the added degree of freedom for the ramp length of the AME weight function and the AME function parameters that were optimized for the task of GIF.

A. Determination of the AME Function Parameters

The AME weight function parameters for the QCP method were optimized by using a grid search to determine the AME parameters that minimized the overall error on a synthetic training set. In detail, the optimization was performed as follows.

First, a training set of synthetic vowels at an 8 kHz sampling rate was generated using the LF model as excitation. 625 different LF pulses were created by interpolating between maximum and minimum average values reported in [27]. F_0 was varied from 75 Hz to 405 Hz with a 10 Hz-increment. The vocal tract was adjusted as in [28], by simulating three different vowels ([a], [u], [i]) using an 8th order all-pole filter i.e. 4 formants. In total, the procedure yielded 63,750 test vowels.

Second, the QCP inverse filtering procedure was applied to the training set by using varying values of DQ, PQ, and N_{ramp}. DQ was varied between 0.4 and 1 with an increment of 0.05, PQ was varied between 0 and 0.2 with an increment of 0.025, N_{ramp} was varied between 0 and 10 samples with an increment of 1, and the alternative RQ parameter was varied between 0 and 0.2 with an increment of 0.05.

Third, the optimal combination of the AME parameters was determined by searching for the minimum of an overall error measure. This overall error measure was a combination of several parametrization schemes that have been used in GIF. The involved error measures were the average absolute relative errors of the Normalized Amplitude Quotient (NAQ) [29], the Quasi-Open Quotient (QOQ) [30], and the Harmonic Richness Factor (HRF) [31], the average absolute error of H1H2 [32], and the Mean-Squared Error (MSE) of the estimation to the reference.

The combined error measure E_{tot} was calculated as follows: For each set of AME parameter combinations, the mean absolute error for each measure type was determined, and they were normalized by the smallest mean error of the respective measure type. Next, the normalized mean errors were summed (with equal proportional weight) to get the total combined error E_{tot}. E_{tot} is a vector, where each cell represents an AME parameter combination, and its value represents the average error for all measures with those parameters; the smaller E_{tot} gets, the better. The theoretical minimum value of E_{tot} is 5, meaning that the minimum value of each measure type is obtained with the same AME parameter combination.

The combined error measures indicated that a fixed N_{ramp} value of 6–8 samples (0.75–1 ms) is the best value for the ramp coefficient, so the value of $N_{ramp} = 7$ samples (0.875 ms) was selected for the rest of the experiments. This means that even though the binary option with $N_{ramp} = 0$ was available, the AME function performs best overall with buffer zones between the open and closed phases, which increases robustness. Fig. 4(a) illustrates the combined error measure E_{tot} with $N_{ramp} = 7$ samples. The parameter combination indices represent different combinations of PQ and DQ, and clear peaks and valleys can be seen for various parameter combinations. A PQ value between 0.05 and 0.1 was the best overall (with 0.05 giving the minimum value). The DQ value of 0.7 gave the overall minimum value, with similar results between 0.6 and 0.85.

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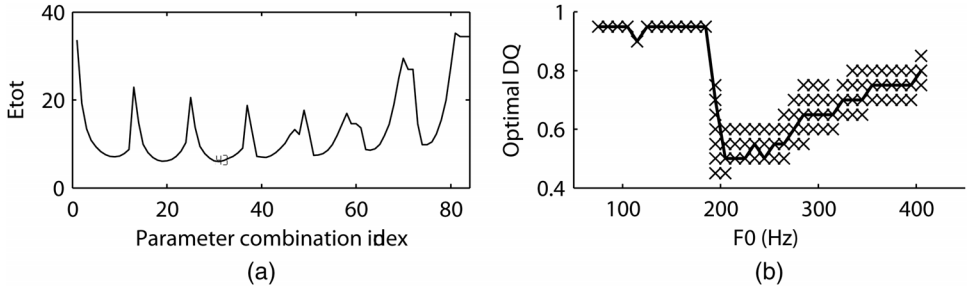

Fig. 4. The overall optimization of AME window parameters (a), and the F_0 adaptive determination of DQ (b). (a) Trimmed ($N_{ramp} = 7$) visual representation of the performed grid search to determine the optimal AME parameters. Minimum index $= 31$ corresponding to $PQ = 0.05$ and $DQ = 0.7$. (b) The optimal DQ parameter (solid line) and the parameters within 10% combined error measure score of the optimal ('x's) for the AME window for each F_0 bin ($PQ = 0.05$, $N_{ramp} = 7$).

By keeping PQ and N_{ramp} fixed ($PQ = 0.05, N_{ramp} = 7$ samples), the dependency of the optimal value of DQ on F_0 was tested next. The optimal DQ was determined for each F_0 bin of the test set and the resulting DQ contour is illustrated in Fig. 4(b). As shown by the contour, it was found that the optimal DQ parameter clearly shifted depending on F_0. For low-pitched signals (i.e. $F_0 < 200$ Hz), the optimal DQ was close to unity indicating that the effect of AME weighting becomes neutralized and the WLP model approaches the conventional LP model. When F_0 rises, however, the DQ contour shows a rapid decrease in vicinity of 200 Hz demonstrating that more accurate vocal tract models are obtained by increasing the attenuation width in the AME weighting function. When F_0 rises further (i.e. $F_0 > 250$ Hz), it can be seen from Fig. 4(b) that the optimal value of DQ starts to grow. This behavior is explained by the use of a fixed N_{ramp} value for all the signals in the experiment. With a fixed N_{ramp}, namely, the AME weighting function has fewer non-attenuated samples for signals of high pitch. If the duration of the non-attenuated part becomes short, the AME function starts to downgrade too much also the contribution of samples that are located outside the instant of glottal closure which results in deterioration of the corresponding vocal tract models. Hence, the growing trend of the DQ contour in Fig. 4(b) at F_0 values after about 250 Hz shows that the optimal AME-function for high-pitched signals is the one that involves attenuation at glottal closure instants (i.e. $DQ < 1$) without overly reducing the duration of the non-attenuated section (i.e. $DQ > 0.6$).

The pitch-adaptive optimization of the parameters was justified because F_0 estimation is relatively reliable in real-world applications, and because the pitch information (the GCIs) is already needed for the correct functioning of the QCP method. Following the same logic, phonation and vowel types were not taken into separate consideration at this stage because their reliable determination is not possible without *a priori* knowledge and/or complex recognition algorithms.

In summary, the experiments conducted in this section resulted in parameter values that are shown for the fixed and pitch-adaptive AME function in Table I. The pitch-adaptive values will be used in the remaining part of the article.

TABLE I
THE OBTAINED OPTIMAL PARAMETERS FOR THE AME WEIGHT FUNCTION.

		PQ	DQ	N_{ramp}
Fixed		0.05	0.7	7
Pitch-adaptive	$F_0 < 190$ Hz	0.05	0.95	7
	$190 \leq F_0 < 280$ Hz	0.05	0.55	7
	$F_0 \geq 280$ Hz	0.05	0.7	7

V. REFERENCE METHODS

The performance of QCP was compared with three previously developed GIF methods: Closed Phase Covariance [3], Iterative Adaptive Inverse Filtering [4], and Complex Cepstrum Decomposition [11]. In addition, WLP analysis with the STE weight function [16] was used as a reference method. All of these four reference methods are described in detail in this section.

A. Closed Phase Covariance Analysis

As discussed in Section IV, the CP covariance [3] inverse filtering method is based on the estimation of the vocal tract transfer function during the closed phase (Fig. 1) of the glottal excitation. The LP analysis is done traditionally by using the covariance criterion instead of the autocorrelation criterion in order to better utilize the sparse data samples located in the closed phase.

The main drawbacks of the CP method are the robust and accurate estimation of the closed phase, as well as the problem of having too short analysis windows for reliable vocal tract estimation. Various improvements have been proposed to the original CP method, including the use of a parallel electroglottography (EGG) signal for more reliable estimation of the closed phase [33], discrete all-pole modeling (DAP, [17]) for more precise envelope detection [11], [34], the use of multiple pitch cycles (for high-pitched voices) to obtain enough data for the vocal tract filter estimation [21], and the use of constrained LP analysis [35].

In this study, the CP method uses the baseline covariance LP with an order $p = 10$, which is suitable to model sufficiently all the formants when speech is sampled at 8 kHz [1]. For synthetic speech, the closed phase is determined straight from the glottal flow derivative waveforms with the help of the pitch information. The minimum allowed length for the closed phase was set to $p + 1$ samples. For high-pitched voices ($F_0 \geq 200$ Hz), two pitch periods were used for the filter estimation [21] to obtain more competitive results for the "High" categories of the sample databases, as one of the main goals of the study was to develop a GIF method that increases performance at high F_0 s.

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B. Iterative Adaptive Inverse Filtering

IAIF [4] is an autocorrelation LP based GIF method that is based on *a priori* knowledge of the overall shape of the vocal tract and glottal source transfer functions. IAIF tries to remove the tilting effect of the glottal source from the speech spectrum with an iterative analysis procedure that is repeated twice. In the first iteration, the contribution of the glottal source is modeled as a first-order IIR filter, the vocal tract transfer function is obtained, and the frame is inverse filtered using this information. An estimate for the glottal source is obtained, and LP analysis of order q is used to obtain its envelope estimate for the second iteration. The benefit of the IAIF algorithm is that it can be easily computed from the input speech waveform without any need for additional parameter estimation.

In this study, the pitch-asynchronous version of the IAIF algorithm was used with the orders $p = 10$ and $q = 4$. The LP analysis at each stage was performed by using the Hanning window.

C. Complex Cepstrum Decomposition

The CCD method [11] and the ZZT inverse filtering method [7] are two different methods that give the same result, with CCD being more computationally efficient. Both of these methods are based on the mixed-phase model of speech [36], which assumes that speech is composed of both minimum-phase (causal) and maximum-phase (anti-causal) components. In the model, the vocal tract impulse response and the glottal return phase are considered as minimum-phase signals, and the open phase of the glottal flow signal is considered as a maximum-phase signal. These signals can be separated via the process of *mixed-phase decomposition*. In the CCD method, the complex cepstrum (CC) of a two pitch-period long, GCI centered, and Blackman windowed frame is computed, and the positive and negative indices of the CC are separated (*liftered*) from each other. The negative/positive indices correspond to the anti-causal/causal components of the signal, and thus the glottal excitation (without the return phase) is obtained by applying the inverse complex cepstrum operation to the liftered signal. In this study, the CCD method was obtained as part of the GLOAT toolbox [37], which is implemented based on [11] by the same author.

D. Weighted Linear Prediction

To gain an understanding on the effect of the AME weight function in the proposed QCP method, WLP with the short-term energy (STE) weight function (Eq. (7)) was selected as the final reference method. The method will be referred to as the WLP method. The WLP method is identical to the proposed QCP method, but instead of the AME weight function, the STE weight function is used in the computation of WLP. The STE function is not expected to perform optimally for GIF purposes, because the function gives higher non-zero weight for the open phase than the AME function. The STE function is computed with the baseline energy window length of $M = 12$, as used in [16].

SWLP [19] was not used in the proposed method because initial experiments indicated that the AME weight function is not particularly applicable with the method.

VI. Experiments

The evaluation of the QCP method was performed by comparing the errors on objective measures obtained from inverse filtered sustained synthetic vowels, as well as by a subjective listening test where the analysis/synthesis quality of a vocoder utilizing GIF was tested. Also, the sensitivity of the proposed method to GCI estimation errors was evaluated.

The objective evaluation of inverse filtering performance was done with NAQ, QOQ, H1H2, HRF, and MSE (see Section IV-A). These parameters describe estimated glottal flows from different perspectives. The NAQ parameter is a measure for the relative length of the glottal closing phase [29]. Its value is expected to indicate estimation errors that change the skewness of the flow pulse such as occurrence of severe formant ripple in the closed phase. QOQ is a measure for the relative open phase duration [30]. Therefore, QOQ is in principle affected by estimation errors that occur both in the opening and closing phase of the glottal cycle. H1H2 and HRF are both frequency domain parametrization methods. Their values are affected in principal by estimation errors that occur at any instant of the glottal cycle. However, since both of these frequency domain quantities measure the spectral decay of the glottal flow pulse waveform, the change in the parameter value reflects mainly to estimation error that affects the closing phase. MSE is not based on any known glottal flow parametrization method but is simply a squared difference of the time-domain waveforms of the reference and estimated glottal pulse derivative frames. Its value is affected by all time instants in the glottal cycle, and especially by errors in those samples that are of largest amplitude (i.e. those occurring during the closing phase).

The errors for NAQ, QOQ, and HRF were reported as the average absolute relative error:

$$E_{\mathrm{NAQ}} = \mathrm{E}\left[\frac{|\mathrm{NAQ}_{ref} - \mathrm{NAQ}_{meas}|}{\mathrm{NAQ}_{ref}}\right] \tag{8}$$

$$E_{\mathrm{QOQ}} = \mathrm{E}\left[\frac{|\mathrm{QOQ}_{ref} - \mathrm{QOQ}_{meas}|}{\mathrm{QOQ}_{ref}}\right] \tag{9}$$

$$E_{\mathrm{HRF}} = \mathrm{E}\left[\frac{|\mathrm{HRF}_{ref} - \mathrm{HRF}_{meas}|}{\mathrm{HRF}_{ref}}\right] \tag{10}$$

and the errors for H1H2 and MSE were reported as the average absolute error:

$$E_{\mathrm{H1H2}} = \mathrm{E}\left[|\mathrm{H1H2}_{ref} - \mathrm{H1H2}_{meas}|\right] \tag{11}$$

$$E_{\mathrm{MSE}} = \mathrm{E}\left[\frac{1}{N}\sum_{n=1}^{N}\left(g_{ref}[n] - g_{meas}[n]\right)^2\right] \tag{12}$$

where E denotes expectation.

The objective error measures were tested on two separate sets of sustained vowels, synthesized based of different models of speech production. The first test set, the LF model test set, was synthesized according to the source-filter model with LF model pulses as excitation and all-pole filters as the vocal tract filter. The second test set, the physical model test set, was synthesized according to a physical model of human speech production [38]. Both test sets had a wide range of F_0 s and phonation types. The use of the physical model is beneficial in the evaluation because,

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unlike the LF model, it is independent from the linear source-filter model used in the design of all GIF methods compared.

A. The LF Model Test Set

The LF model is a parametric model for the glottal flow derivative waveform [12], where the t_e, t_p, t_a, and E_e constitute the LF parameters that define the waveform. The LF parameters can be represented in an alternate dimensionless form of E_e, R_a, R_g, and R_k with the advantage that they can be interpolated within their respective ranges obtained from [27] to get a thorough range of possible excitations. The experiments were done on a similar database of synthesized vowels as in the analysis-by-synthesis optimization scheme of Section IV-A. F_0 was varied from 80 Hz to 400 Hz with a 10 Hz-increment. The vocal tract was adjusted as in [28], by simulating three different vowels ([e], [o], [ae]) with an all-pole filter with 8 poles (resulting in 4 formants). The LF test contained a total of 25,344 test vowels.

Particular attention was given to the fact that the database used for the testing was synthesized with different parameter values from the one used for the optimization of the proposed method. The F_0 parameter varied between 80–400 Hz with 10 Hz-increments, and the LF parameters were given four linearly spaced values each, interlaced with the optimization set. The synthesized vowels were [e], [o], and [ae], yielding a total of 25,344 test vowels.

The test samples were divided into two categories based on their pitch: The 'Low' ($F_0 \leq 200$ Hz) and 'High' ($F_0 > 200$ Hz) categories. This was done to gain better insight of the tested methods' performance for various voices.

Even though a wide range of F_0 values and phonation types were involved in the LF model test set, it is worth noting that using these data in evaluation of GIF warrants for caution because sound synthesis is based on simple linear modeling.

B. The Physical Model Test Set

In order not to use test material that is entirely based on linear source-filter modeling, the GIF methods were also evaluated using physical modeling of the speech production mechanism. This approach is different from the one where synthetic speech excited by an artificial form of the glottal excitation is used, because the glottal flow waveform results from the interaction of the self-sustained oscillation of the vocal folds with subglottal and supraglottal pressures, as would occur during real speech production. Hence, the glottal flow waveform generated by this model is expected to provide a more stringent and realistic test of the inverse filtering method than would be permitted by a parametric flow waveform model where no source-tract interaction is incorporated.

In this approach, a computational model of the speech production system was used to generate vowels representative of an adult male, adult female, and an approximately five-year-old child. The voice source component used consisted of a kinematic representation of the medial surface of the vocal folds [39], [40] for which the F_0, surface bulging, adduction, length, and thickness are control parameters. As the vocal fold surfaces are set into vibration the model produces a glottal area signal that is coupled to the acoustic pressures and air flows in the trachea and vocal tract through aerodynamic and acoustic considerations [41]. The resulting glottal flow was determined by the interaction of the glottal area with the time-varying pressures present just inferior and superior to the glottis.

The vocal tract shape was specified by an area function representative of the vowels [a], [i], and [æ]. The acoustic wave propagation in the subglottal and supraglottal airspaces was computed with a wave-reflection model that included energy losses due to yielding walls, viscosity, heat conduction, and radiation at the lips [38]. Detailed parameter values used for the physical model are presented in [20].

Each of the vowels for the male, the female, and the child were generated with eight F_0 values, ranging from 100 Hz to 450 Hz in 50 Hz-increments. Although the full range of these F_0 s is unlikely to be produced by either the male, female, or the child, conducting the experiment with the entire range was desirable for ease in comparison. Vowel duration was 0.4 seconds and F_0 was maintained constant during the utterance. Sampling frequency was finally set to 8 kHz.

C. GCI Estimation Error Sensitivity

The vulnerability of the proposed QCP method to GCI estimation errors was tested by three tests: First, by adding uniformly distributed random noise (in samples) of varying deviation to the correct GCIs. Second, by adding or removing various numbers of detected GCIs, and third, by adding uniformly distributed, biased random noise to the correct GCIs. The LF model test set was used as reference because of the larger volume of data, and the error measures were computed, with reference values provided by the GCI errorless QCP method and the WLP method.

For the additive noise, the tested deviations *dev* were 8 samples (error ranging from -4 to $+4$ samples) and 16 samples (error ranging from -8 to $+8$ samples). An error was randomly assigned for each GCI independently from other GCIs, and this was repeated ten times for each test frame to get the average values for all errors. This test can be thought of as particularly harsh to the QCP method, because in reality rarely *all* detected GCIs have significant errors in them.

For the case of the missing or extra GCIs, two GCIs were added or removed from the correct GCI information vectors. In the removed cases, the minimum number of GCIs could not be smaller than 1, and the order of the removed GCIs was randomized. In the cases of extra GCIs, the positions of the additive GCIs were randomized within the analysis frame so that they would not overlap with already existing GCIs.

For the effect of the error bias, the range of the tested deviations was the same as in the first test, but their center points were categorized into three different groups: left-biased, non-biased, and right-biased. The ranges of the left- and right-biased groups' deviations were $[-dev, 0]$ and $[0, dev]$, respectively. As in the first test, the non-biased group's deviation was $[-dev/2, dev/2]$. Due to the space limitation of this article, only the NAQ parameter was considered in the third test. The deviation values were however computed with more density, within the interval $dev = 0 : 16$.

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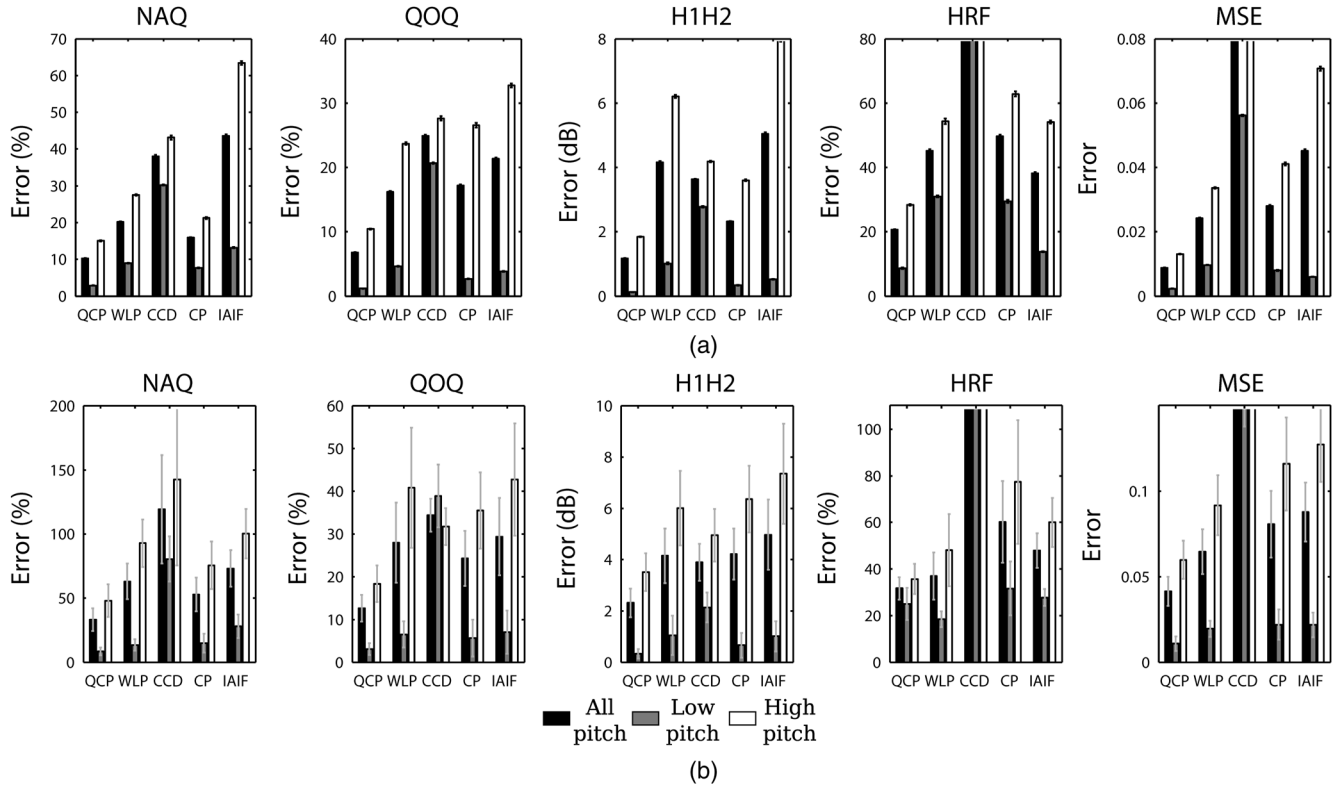

Fig. 5. The average error measures with their 95% confidence intervals on the selected glottal source parameters for the competing GIF methods shown for the LF model test (a) and the physical model test (b) . 'Low' group consists of samples with $F_0 \leq 200$ Hz, and the 'High' group consists of samples with $F_0 > 200$ Hz.

TABLE II
AVERAGE ERRORS OF THE GLOTTAL SOURCE PARAMETERS FOR THE LF AND PHYSICAL MODEL TEST SETS. GIF METHODS WITH THE BEST RESULTS ARE PRINTED WITH BOLD FONT

	NAQ			QOQ			H1H2 (dB)			HRF			MSE		
LF model test set	All	Low	High	All	Low	High	All	Low	High	All	Low	High	All	Low	High
QCP	**0.10**	**0.03**	**0.16**	**0.07**	**0.02**	**0.12**	**1.2**	**0.2**	**2.1**	**0.19**	**0.07**	**0.30**	**0.009**	**0.002**	**0.013**
WLP	0.20	0.10	0.30	0.16	0.06	0.26	4.2	1.3	6.8	0.45	0.31	0.58	0.024	0.010	0.034
CCD	0.38	0.30	0.43	0.25	0.21	0.27	3.6	2.8	4.2	1.9	2.0	1.8	0.096	0.056	0.122
CP	0.16	0.09	0.23	0.17	0.05	0.29	2.3	0.6	3.9	0.50	0.29	0.69	0.028	0.008	0.041
IAIF	0.44	0.18	0.68	0.21	0.07	0.35	5.0	0.9	8.9	0.38	0.16	0.59	0.045	0.006	0.071
Physical model test set	All	Low	High	All	Low	High	All	Low	High	All	Low	High	All	Low	High
QCP	**0.33**	**0.08**	**0.48**	**0.13**	**0.03**	**0.18**	**2.3**	**0.3**	**3.5**	**0.32**	0.25	**0.36**	**0.042**	**0.011**	**0.060**
WLP	0.63	0.13	0.92	0.28	0.07	0.41	4.2	1.0	6.0	0.37	**0.18**	0.48	0.065	0.020	0.092
CCD	1.19	0.81	1.43	0.34	0.39	0.32	3.9	2.1	5.0	$\gg 1$	$\gg 1$	$\gg 1$	0.199	0.165	0.220
CP	0.52	0.15	0.76	0.24	0.06	0.36	4.2	0.7	6.4	0.60	0.32	0.77	0.081	0.022	0.116
IAIF	0.73	0.28	1.00	0.29	0.07	0.43	5.0	1.0	7.4	0.48	0.28	0.60	0.088	0.022	0.127

D. The Subjective Listening Test on Analysis/Synthesis Quality

The performance of the QCP method was tested in the physiologically oriented vocoder GlottHMM [14]. GlottHMM utilizes the IAIF method to decompose speech into the glottal source signal and the vocal tract filter using conventional LP. For the purpose of this study, the conventional LP inside the IAIF method was replaced with the QCP method in order to reduce the biasing effect of the harmonics on the spectral envelope. The analysis-synthesis quality of the baseline LP-based vocoder was compared to the QCP-modified vocoder in an ABX listening test. In the test, listeners were presented with a natural reference sample and two vocoded samples. The task of the listener was to select the one that sounded more like the reference sample (or to express no preference). The test consisted of a total of 20 samples of normal and shouted speech by male and female speakers. A total of 15 native Finnish listeners assessed the performance of the methods.

VII. RESULTS

A. The LF Model Test Set

The results for the LF model test set are presented in Fig. 5a and in numerical form in Table II. It can be seen that the QCP method outperforms the compared methods clearly in all categories and measures. The performance has a slightly higher relative improvement in the 'Low' category than in the 'High' category. The QCP method also shows a clear improvement over the WLP method with the STE function, which scored slightly higher errors than the second-best rated CP method.

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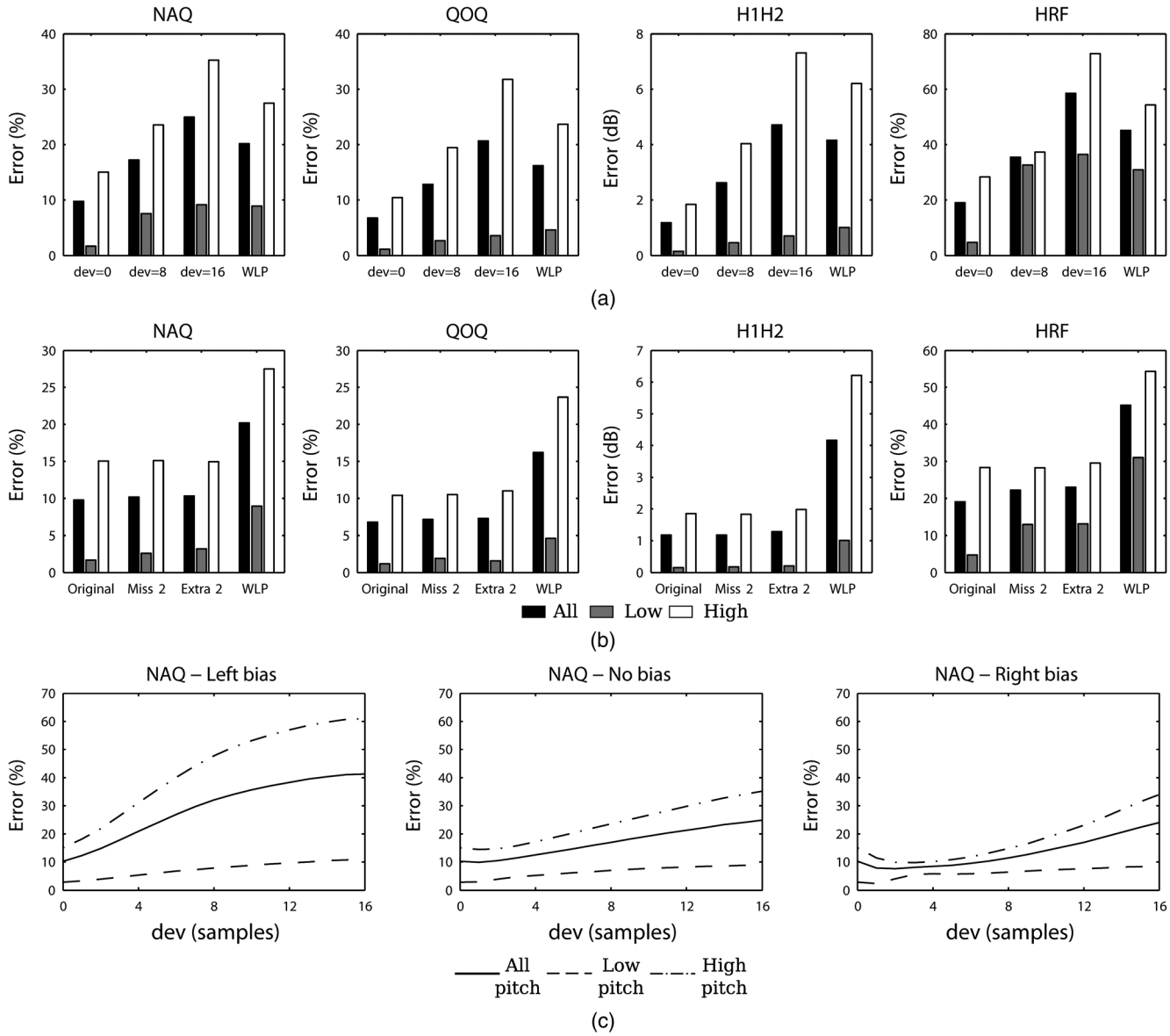

Fig. 6. Sensitivity of the proposed QCP method to GCI estimation errors. WLP with the STE weight function is shown as a reference. (a) Error measures for the proposed method with uniformly distributed additive noise with a deviation of *dev* samples in each GCI. (b) Error measures for the proposed method with missing or extra GCIs. (c) Error measures for the NAQ parameter for left-biased, non-biased and right-biased GCI error deviations (*dev*).

Also to be noted is the poor performance of the CCD method, which is different from the previously reported results for LF-model based test samples [11]. The low results in the present study might have been caused by the fact that the obtained version of the CCD method estimates only the opening and closing phases of a glottal period until the GCI, discarding the return phase completely (see Fig. 1). This leads to systematical errors in the glottal flow parameter estimations.

B. The Physical Model Test Set

The results for the physical model test set are presented in Fig. 5b and in Table II. The use of the physical model test set yielded similar results to the ones obtained with the LF model test set. The QCP method performed best in all categories except for the 'Low' HRF measure. The average error measures were however higher for the physical model test set than for the LF model test set. Also, the fixed and pitch-adaptive versions of the QCP method scored similar results with the physical model test set, which suggests that the optimal DQ parameter trends depicted in Fig. 4(b) are relevant mainly with the LF model.

C. GCI Estimation Error Sensitivity

The sensitivity of the QCP method to GCI estimation errors for the case of randomly shifted GCIs is presented in Fig. 6(a). It can be seen that the error is almost a linearly increasing function of *dev* for all parameters. Also to be noted is that the high-pitched sample set is affected most by the growing GCI position error. This means that, on average, the performance of the QCP method deteriorates more for high-pitched voices than for low-pitched voices in the case of real speech where accurate GCI information is not available.

The effect of missing or extra GCIs to the performance of QCP is presented on Fig. 6(b). Surprisingly, the results indicate that missing or extra GCIs do not greatly affect the results (assuming that the detected GCIs are accurate and there is at least one detected).

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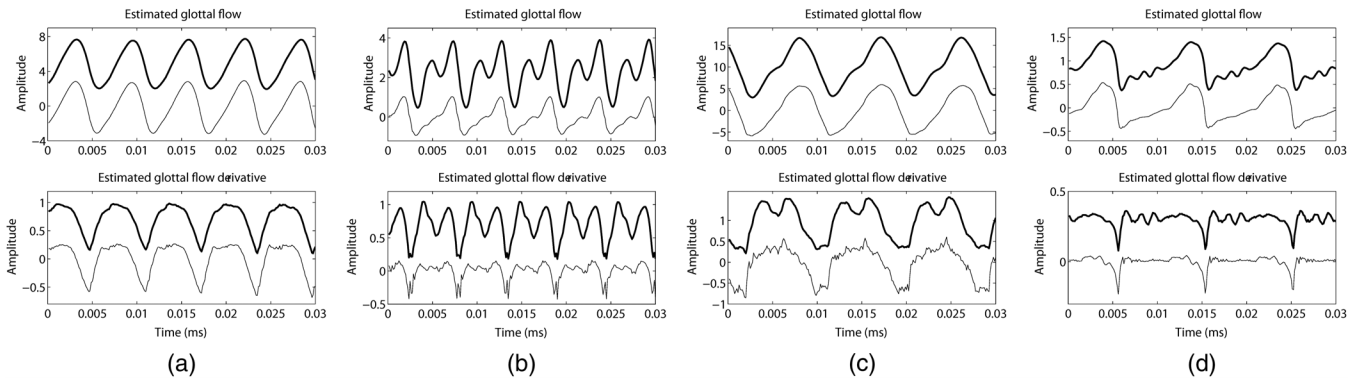

Fig. 7. Inverse filtering examples for real speech. The bold line is the inverse filtering result obtained by the IAIF method, and the thin line is the inverse filtering result computed by the QCP method. The IAIF plot has been raised for image clarity. (a) [i] vowel, soft female voice. (b) [y] vowel, pressed female voice. (c) [i] vowel, soft male voice. (d) [a] vowel, pressed male voice.

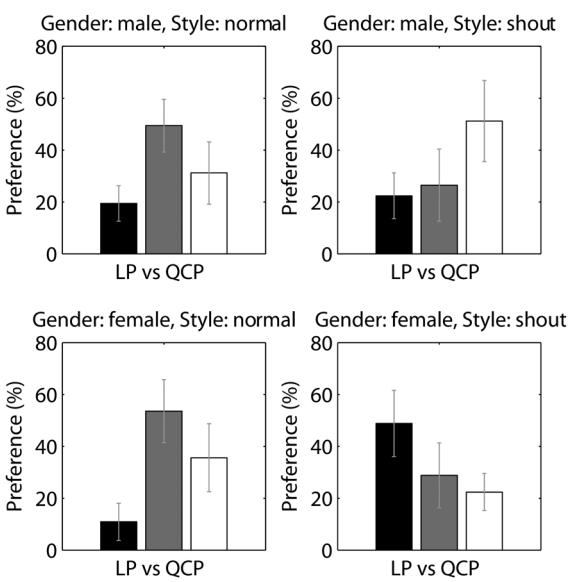

Fig. 8. Subjective listening test results for different voice types and their 95% confidence intervals. The grey bars indicate the results for no preference.

The bias on the GCI error is seen to have a noticeable effect on the results, as depicted by Fig. 6c. Left-biased GCI estimates degrade the results more than right-biased GCI estimates, especially for high-pitched vowels. This is expected, because bias to the left moves the emphasized portion of the AME weighting function to the vicinity of the GCI, the instant of maximum excitation. The reason why high-pitched vowels are affected more is that a sample-fixed GCI error introduces more proportional error to the creation of the AME weight function at higher F_0s.

D. The Subjective Listening Test

The results of the subjective listening test are presented in Fig. 8. It is noticeable that the QCP method improves the perceived analysis/synthesis quality of normal speech for male and female speakers, but the results for shouted speech are not as good for the QCP method, especially for females. This is probably caused by GCI estimation errors that resulted in audible clipping artifacts on the resynthesized speech waveform. Shouted speech has the highest F_0 range and also the most challenging waveforms, caused by rapid changes of the excitation, which can make the assumption of sustained behavior inside an analysis frame inadequate. This in turn makes feature extraction unreliable. This interpretation would align with the findings of the GCI error test, where it was indicated that the performance of the QCP method deteriorates significantly for high-pitched speech when GCI position errors are present.

E. A Representative Example on Real Speech

The GIF performance of the QCP method is illustrated in Fig. 7 where various sustained vowels from female and male speakers are inverse filtered with QCP and IAIF. The QCP method can be seen to attenuate the fluctuations of the closed phase of the glottal flow more than the IAIF method in most cases. Also, the glottal open and closed phases are easier to detect from the glottal flow derivative waveforms in the examples.

VIII. Conclusion

A new glottal inverse filtering technique, QCP analysis, was introduced in this article. QCP analysis performs a closed phase type analysis over a time frame of multiple fundamental periods by utilizing WLP. A specific weight function attenuates the contribution of the (quasi) open phase, resulting in an estimate of the vocal tract transfer function that is less influenced by the excitation. The method was evaluated with objective measures against well-known GIF methods where the objective measures were obtained by inverse filtering synthetic vowel databases, constructed via the conventional LF model as well as via physical modeling of human speech production. The robustness of the method with regards to its dependency on GCI detection was evaluated with objective measures. Finally, a subjective listening test was performed to measure the analysis/synthesis performance of the proposed method in a physiologically oriented vocoder.

The results indicate that the proposed method outperforms the other four GIF methods in the vast majority of cases in both test sets and with low- and high-pitched vowels. There are two main limitations to the proposed method. First, the method requires information about the GCIs to form the appropriate WLP weight function, which might cause degradation in its performance in real-world situations where accurate GCI data is not available. This is well illustrated by both the objective measure experiments as well as the subjective listening test, where the

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performance of the QCP method deteriorates as the estimated GCIs become more erroneous. Second, the proposed method does not guarantee filter stability, which causes an additional computational cost in GIF applications were all-pole synthesis is needed.

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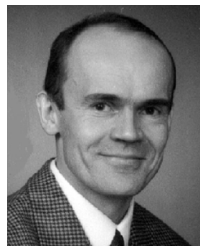

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